Cole D. Butler, PhD

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Current Position

University of Georgia, Department of Entomology, Postdoctoral Research Associate in the lab of Dr. Bill Snyder — 2025 – Present

Education

PhD **North Carolina State University**, Major: Biomathematics; Minors: Statistics; Interdisciplinary Perspectives on Genes and Genomes — 2020 – 2025

- GPA: 4.0/4.0; Genetics and Genomics Scholar
- NSF Graduate Research Fellow, 2022-2025
- Dissertation: The Role of Genetics in Pest Population Control

BS **University of Maine**, Mathematics, *summa cum laude* — 2016 – 2020

- GPA: 3.95/4.0; Honors College (highest honors); Dean's List all semesters

Publications

- **Butler, C.**, Dhole, S., Gould, F., and A. Lloyd. Partially functional resistance interferes with gene drive. *In progress.*
- **Butler, C.** and A. Lloyd (2026). Spatial confinement of gene drives: Assessing risk of failure using global sensitivity analysis. *Evol Appl* (in review). Preprint DOI: 10.64898/2026.02.18.706608.
- Baltzegar, J., **Butler, C.**, Ding, J., Beeson, C., Reed, E., Reiskind, M., and M. Burford Reiskind (2026). Emergence and spatial distribution of knockdown resistance in a suburban population of *Aedes albopictus*. *Parasit Vectors*. 19(1); 409.
- **Butler, C.**, Ding, J., Baltzegar, J., Brown, Z., Burford Reiskind, M., and M. Reiskind (2026). Socioeconomic predictors of knockdown resistance in the globally invasive tiger mosquito *Aedes albopictus*. *R Soc Open Sci*. 13(7): 251678.
- **Butler, C.** and A. Lloyd (2025). How population control of pests is modulated by density dependence: The perspective of genetic biocontrol. *J Theor Biol*. 614; 112215.
- Yadav, A., **Butler, C.**, Yamamoto, A., Patil, A., Lloyd, A., and M. Scott (2023). CRISPR-Cas9 based split homing gene-drive targeting *doublesex* for population suppression of the global fruit pest *Drosophila suzukii*. *Proc Natl Acad Sci U S A*. 120(25); e2301525120. **Received 2023 GGA Paper of the Year Award**
- **Butler, C.** and P. Stechlinski (2023). Modeling opioid abuse: A case study of the opioid crisis in New England. *Bull Math Biol*. 85(6); 45.
- **Butler, C.**, Cheng, J., Correa, L., Preciado, M., Rios, A., Montalvo, C., and C. Kribs (2021). Comparison of screening for methicillin-resistant *Staphylococcus aureus* (MRSA) at hospital admission and discharge. *Letters in Biomath.* 8(1).

Invited Talks

- **Butler, C.** "Remote sensing to detect crop damage by an invasive species," 2026 Biology and Medicine Through Mathematics Conference, Virginia Commonwealth University, in-person talk
- **Butler, C.**, Ding, J., Baltzegar, J., Brown, Z., Burford Reiskind, M., and Reiskind, M. "Socioeconomic Drivers of Knockdown Resistance in the Invasive Tiger Mosquito," 2025 Genetics and Genomics Academy Retreat, NC State University, in-person talk

- **Butler, C.** “Mathematical modeling and current research on the sweetpotato whitefly (*Bemisia tabaci*),” 2025 Biomathematics Seminar, NC State University, virtual talk
- **Butler, C.** "Modeling insect populations and the next generation of genetic pest control," 2023 Mathematics Colloquium, University of Maine, in-person talk
- **Butler, C.** "The importance of dispersal in gene drive control," presented at the following venues:
  - 2023 Joint Mathematics Meeting, in-person talk;
  - 2023 Biomathematics Seminar, Texas Tech University, virtual talk

Selected Grants and Fellowships

- Co-PI, NSF PRFB, 2026, *Untangling Mosquito Ecology: Landscape Drivers of Mosquito Communities* (Pending) ($220,000)
- Co-PI, USDA APHIS PPA 7721, 2026, *Remote Sensing of Two-Spot Cotton Leafhopper Damage* (Pending) ($114,907)
- Co-PI, USDA NIFA AFRI EWD, 2026, *From Pixels to Pests: A Data-Driven Approach to Leafhopper Prediction and Monitoring* (Pending) ($194,480)
- Co-PI, GGA Seed Grant, 2024, *Evaluating genetic and socio-economic factors affecting insecticide resistance in a population of* Aedes albopictus *from Wake County, NC* ($25,000)
- Co-PI, GOHA Seed Grant, 2024, *One Health Implications of Insecticide Resistance in Pathogen Vectors* ($35,096)
- NSF Graduate Research Fellowship, 2022-2025 ($138,000)

Research Experience

Modeling pest abundance and crop damage in the Southeastern U.S. — Aug 2025 – Present

- Developed a remote sensing pipeline to detect damage caused by the invasive two-spot leafhopper (*Amrasca biguttula*) in cotton
- Using statistical models to identify predictors of silverleaf whitefly (*Bemisia tabaci*) abundance and investigate how drought modifies trophic interactions

Using mathematical modeling to evaluate drawbacks in genetic biocontrol — Aug 2025 – Present

- Conducted equilibrium and stability analyses to understand how organism life history affects genetic biocontrol efficacy (advised by Dr. Alun Lloyd)
- Analyzed spatial dynamics of gene drive spread and quantified risk of failure under variable conditions (advised by Dr. Alun Lloyd)
- Developed models of gene drive resistance to address overlooked problems in current approaches (collab. w/Drs. Alun Lloyd, Fred Gould, and Sumit Dhole)

Modeling genetic biocontrol of spotted wing Drosophila — Jan 2022 – July 2025

- Developed data-driven mathematical models with molecular biologists to evaluate genetic biocontrol technologies in spotted wing Drosophila (*Drosophila suzukii*) (collab. w/Dr. Max Scott)
- Simulated experimental designs for cage trials and technology development

Identifying socioeconomic predictors of insecticide resistance in mosquitoes — Mar 2023 – July 2025

- Planned and oversaw field collections of the tiger mosquito (*Aedes albopictus*) in Raleigh, NC (collab. w/Drs. Michael Reiskind, Martha Burford Reiskind, Jen Baltzegar, and Zack Brown)
- Used statistical models to analyze associations between insecticide resistance frequency and socioeconomic predictors